

MATA37H3 — Week 12

Answer Key & Misconception Notes

For tutor use only — do not show the student until after their attempt. (Power Series II: Taylor Polynomials)

EXAMPLE 1 — Answer

$p_3(x) = 2x^3 - x^2 + 5$ (equals f itself, since f has degree 3). $p_2(x) = -x^2 + 5$ (drop the x^3 term). $p_1(x) = 5$ (drop x^3 and x^2 terms — **not** $5 + 0 \cdot x$ written differently, just the constant).
Common misconception: Trying to differentiate f from scratch instead of reading the coefficients off directly — wastes time and invites arithmetic slips; uniqueness makes this a zero-computation problem.

EXAMPLE 2 — Answer

With $u = x + 1$ (so $x = u - 1$): $x^3 = (u - 1)^3 = u^3 - 3u^2 + 3u - 1$, so $p_3(x) = -1 + 3(x + 1) - 3(x + 1)^2 + (x + 1)^3$. Check via coefficient formula: $f(-1) = -1$, $f'(-1) = 3$, $f''(-1) = -6 \Rightarrow \frac{-6}{2!} = -3$, $f'''(-1) = 6 \Rightarrow \frac{6}{3!} = 1$ — matches.

Common misconception: Sign errors expanding $(u - 1)^3$ — dropping the alternating signs on the odd-power terms.

EXAMPLE 3 — Answer

$Q(x) = 1 + 0(x - 2) + \frac{-6}{2!}(x - 2)^2 + 0(x - 2)^3 + \frac{24}{4!}(x - 2)^4 = 1 - 3(x - 2)^2 + (x - 2)^4$.

Common misconception: Forgetting to include the *zero* coefficients explicitly when reasoning through the pattern — students sometimes skip straight to only the nonzero terms and lose track of which power goes with which given derivative.

EXAMPLE 4 — Answer

$p_3(x) = 1 + x + \frac{1}{2!}x^2 + \frac{1}{3!}x^3 = 1 + x + \frac{x^2}{2} + \frac{x^3}{6}$.

Common misconception: Forgetting to divide by $k!$ at each step — writing $1 + x + x^2 + x^3$ (using the derivative values directly as coefficients) instead of dividing each by the correct factorial.

EXAMPLE 5 — Answer

$p_2(x) = 1 + 0 \cdot x + \frac{-1}{2!}x^2 = 1 - \frac{x^2}{2}$.

Common misconception: Forgetting the negative sign on $f''(0) = -1$, giving $1 + \frac{x^2}{2}$ instead of $1 - \frac{x^2}{2}$ — a sign slip that silently produces the wrong concavity in the approximation.

EXAMPLE 6 — Answer

Both methods compute the **same** degree- ≤ 4 polynomial matching f and its derivatives up to order 4 at $a = 0$; Theorem 6.5 (Uniqueness of Power Series) guarantees only one such polynomial exists, so any correct method must land on it: $p_4(x) = x^4$ (trivially, since f already has degree 4).

Common misconception: Treating “two different methods might give two different but both “correct” answers” as plausible — uniqueness rules this out completely; if two computations of a Taylor polynomial disagree, at least one has an arithmetic error.

EXAMPLE 7 — Answer

True. If $\deg f = 7 \leq 10$, then f ’s own expansion already has zero coefficients for every power above 7, so p_{10} just reproduces f exactly (Taylor polynomials of degree $\geq \deg f$ always equal f exactly, with no remainder). It’s p_n for $n < \deg f$ that would truncate real information.

Common misconception: Reflexively assuming “higher-degree Taylor polynomial = better approximation, never exact” — for a polynomial f , any p_n with $n \geq \deg f$ is not an approximation at all, it’s f itself.

EXAMPLE 8 — Answer

Truncating in powers of x only agrees with truncating in powers of $(x - a)$ when $a = 0$ — for a nonzero center, expanding f in powers of $(x - a)$ redistributes coefficients across all powers (via the binomial expansion of each $x^k = ((x - a) + a)^k$), so “zeroing out high powers of x ” and “zeroing out high powers of $(x - a)$ ” generally keep or discard **different** information about f near a .

Common misconception: Assuming truncation is basis-independent — which power’s coefficients you zero out depends entirely on which variable (x or $x - a$) the expansion is written in; the Taylor polynomial construction always uses $(x - a)$, centered at the point of interest.